

CS & IT ENGINEERING

Programming in C

Functions and Storage Classes

Lec-05



By- Pankaj Sharma sir



TOPICS TO BE
COVERED

Recursion

Q.1

```
void f(int n)
{
    if(n<=0)
        return;
    printf("%d",n);
    f(n-1);
}
```

What is the output of f(5)



Q.2

```
void f(int n)
{
    if(n<=0)
        return;
    f(n-1);
    printf("%d",n);
}
```

What is the output of f(5)



Q.3

```
void f(int n)
{
    if(n<=0)
        return;
    f(n-1);
    printf("%d",n);
    f(n-1);
}
```

What is the output of f(4)



Q.4

```
int f(int n)
{
    if(n<=1)
        return n;
    return f(n/2) + f(n/2) + 1;
}
```

What is the output of f(5)

Q.5

```
int f(int n)
{
    if(n<=1)
        return n;
    return f(n/2) + n/2 ;
}
```

What is the output of f(12)



Q.6

```
int f(int n)
```

```
{
```

```
    if(n<=1)
```

```
        return n;
```

```
    if(n%2)
```

```
        return f(n/2) + n;
```

```
    return f(n/3) + n;
```

```
}
```

output of f(22) ?





Q.7

Consider the code :

```
/* Assume that n>=0 */  
void fun(int n)  
{  
    if(n==0)  
        return 0;  
    fun(n/2);  
    printf("%d",n%2);  
}
```

output of f(11)?

Q.8



Consider the following C program :

```
void foo(int n , int sum) {
```

```
    int k=0,j=0;
```

```
    if(n==0)
```

```
        return;
```

```
    k=n%10;
```

```
    j=n/10;
```

```
    sum=sum + k;
```

```
    foo(j,sum);
```

```
    printf("%d",k);
```

```
}
```

```
void main(){
```

```
    int a=2018,sum=0;
```

```
    foo(a,sum);
```

```
    printf("%d",sum);
```

```
}
```

Output?

A.

8, 4, 0, 2, 14

B.

8,4,0,2,0

C.

2,0,4,8,14

D.

2,0,4,8,0

Q.9

```
void main()
{
    static int var=5;
    printf("%d",var--);
    if(var)
        main();
}
```



Q.10

```
void main()
{
    static int i =5;
    if(--i)
    {
        main();
        printf("%d",i);
    }
}
```



Q.11



predict the output

```
int fun(int x)
{
    if(x%2==0)
        return fun(fun(x-1));
    else
        return(x++);
}

int main()
{
    printf("%d",f(12));
    getchar();
    return 0;
}
```

A.

10

B.

11

C.

12

D.

None of these

Q.12



```
int fun(int a,int b)
{
    if(b==0)
        return 0;
    if(b%2==0)
        return fun(a+a,b/2);
    return fun(a+a,b/2) + a;
}
```

```
int main()
{
    printf("%d",fun(4,3));
    getchar();
    return 0;
}
```

A.

12

B.

81

C.

64

D.

8

Q.13



Consider the following C function :

```
int f(int n)
```

```
{
```

```
    static int r=0;
```

```
    if(n<=0)
```

```
        return 1;
```

```
    if(n<3)
```

```
    {
```

```
        r=n;
```

```
        return f(n-2) + 2;
```

```
    }
```

```
    return f(n-1) + r;
```

```
}
```

what is the value of f(5)

A.

5

B.

7

C.

9

D.

18

Q.14



Consider the following recursive C function
unsigned int foo(unsigned int n, unsigned int r)

```
{  
    if(n>0)  
        return (n%r) + foo(n/r, r);  
    else  
        return 0;  
}
```

output of foo(513,2)

A.

9

B.

8

C.

5

D.

2

Q.15

Which of the following statements is/are valid?



A.

`return a+b;`

B.

`return a,b,c;`

C.

`return (a,b,c);`

D.

All of them

Q.16

```
int fun(int x)
{
    if(x>3)
        return fun(x-4) + fun(x-1) + 1;
    return 1;
}
```

Find the value returned by fun(12)



Q.17

Predict output of following program

```
#include <stdio.h>
```

```
int fun(int n)
```

```
{
```

```
    if (n == 4)
```

```
        return n;
```

```
    else return 2*fun(n+1);
```

```
}
```

```
int main()
```

```
{
```

```
    printf("%d ", fun(2));
```

```
    return 0;
```

```
}
```

A.

4

B.

8

C.

16

D.

error



Q.18

Consider the following recursive function $\text{fun}(x, y)$. What is the value of $\text{fun}(4, 3)$

```
int fun(int x, int y)
{
    if (x == 0)
        return y;
    return fun(x - 1, x + y);
}
```

A. 13

B. 12

C. 9

D. 10



Q.19

What does the following function do?

```
int fun(int x, int y)
{
    if (y == 0) return 0;
    return (x + fun(x, y-1));
}
```

A.

$x + y$

B.

$x + x * y$

C.

$x * y$

D.

$\text{pow}(x, y)$



Q.20

What does fun2() do in general?

```
int fun(int x, int y)
{
    if (y == 0) return 0;
    return (x + fun(x, y-1));
}
```

```
int fun2(int a, int b)
{
    if (b == 0) return 1;
    return fun(a, fun2(a, b-1));
}
```

A.

$x * y$

B.

$x + x * y$

C.

$\text{pow}(x, y)$

D.

$\text{pow}(y, x)$



Q.21

Output of following program?

```
#include<stdio.h>
```

```
void print(int n){
```

```
    if (n > 4000)
```

```
        return;
```

```
    printf("%d ", n);
```

```
    print(2*n);
```

```
    printf("%d ", n);
```

```
}
```

```
int main()
```

```
{
```

```
    print(1000);
```

```
    getchar();
```

```
    return 0;
```

```
}
```

A.

1000 2000 4000

B.

1000 2000 4000 4000 2000 1000

C.

1000 2000 4000 2000 1000

D.

1000 2000 2000 1000

Q.22

What does the following function do?

```
int fun(unsigned int n)
```

```
{
```

```
    if (n == 0 || n == 1)
```

```
        return n;
```

```
    if (n%3 != 0)
```

```
        return 0;
```

```
    return fun(n/3);
```

```
}
```

- A. It returns 1 when n is a multiple of 3, otherwise returns 0
- B. It returns 1 when n is a power of 3, otherwise returns 0
- C. It returns 0 when n is a multiple of 3, otherwise returns 1
- D. It returns 0 when n is a power of 3, otherwise returns 1

Q.23

Predict the output of following program

```
#include <stdio.h>
```

```
int f(int n)
```

```
{
```

```
    if(n <= 1)
```

```
        return 1;
```

```
    if(n%2 == 0)
```

```
        return f(n/2);
```

```
    return f(n/2) + f(n/2+1);
```

```
}
```

```
int main()
```

```
{
```

```
    printf("%d", f(11));
```

```
    return 0;
```

```
}
```

A.

Stack Overflow

B.

3

C.

4

D.

5

Q.24

Consider the following C function:

```
int f(int n)
{
    static int i = 1;
    if (n >= 5)
        return n;
    n = n+i;
    i++;
    return f(n);
}
```

A.

5

B.

6

C.

7

D.

8

The value returned by f(1) is

Q.25



Consider the following C function.

```
int fun (int n)
{
    int x=1, k;
    if (n==1) return x;
    for (k=1; k<5n; ++k)
        x = x + fun(k) * fun(n - k);
    return x;
}
```

A. 0

B. 26

C. 51

D. 71

$$x = 1 + \text{fun}(1) \times \text{fun}(4) + \text{fun}(2) \times \text{fun}(3) \\ + \text{fun}(3) \times \text{fun}(2) + \text{fun}(4) \times \text{fun}(1)$$

The return value of fun(5) is _____.

Q.25

Consider the following C function.

```
int fun (int n)
```

```
{
```

```
    int x=1, k;
```

```
    if (n==1) return x;
```

```
    for (k=1; k<n; ++k)
```

```
        x = x + fun(k) * fun(n - k);
```

```
    return x;
```

```
}
```

The return value of fun(5) is _____.

A.

0

C.

51

B.

26

D.

71

$$fun(1) = 1$$

$$fun(2) = 2$$

$$fun(3) = 5$$

$$fun(4) = 15$$

$$fun(3)$$

$$x = 1 + fun(1) \times fun(2) + fun(2) \times fun(1) \\ = 1 + 1 \times 2 + 2 \times 1$$

$$= 1 + 2 + 2 = 5$$

$$\begin{aligned} fun(4) &= 1 + fun(1) \times fun(3) + fun(2) \times fun(2) + fun(3) \times fun(1) \\ &= 1 + 1 \times 5 + 2 \times 2 + 5 \times 1 \\ &= 1 + 5 + 4 + 5 = 15 \end{aligned}$$

Q.25

Consider the following C function.

```
int fun (int n)
```

```
{
```

```
    int x=1, k;
```

```
    if (n==1) return x;
```

```
    for (k=1; k<n; ++k)
```

```
        x = x + fun(k) * fun(n - k);
```

```
    return x;
```

```
}
```

The return value of fun(5) is 51.

A.

0

C.

51

B.

26

D.

71

$$\text{fun}(1) = 1$$

$$\text{fun}(2) = 2$$

$$\text{fun}(3) = 5$$

$$\text{fun}(4) = 15$$

$$\text{fun}(3)$$

$$x = 1 + \text{fun}(1) \times \text{fun}(2) + \text{fun}(2) \times \text{fun}(1)$$
$$1 + 1 \times 2 + 2 \times 1$$

$$\begin{aligned} \text{fun}(5) &= 1 + \text{fun}(1) \times \text{fun}(4) + \text{fun}(2) \times \text{fun}(3) + \text{fun}(3) \times \text{fun}(2) + \text{fun}(4) \times \text{fun}(1) \\ &= 1 + 15 + 2 \times 5 + 5 \times 2 + 15 \times 1 \\ &= \text{51} \end{aligned}$$

Q.26



Consider the following recursive C function. If `get(6)` function is being called in `main()` then how many times will the `get()` function be invoked before returning to the `main()`?

```
void get (int n)
{
    if (n < 1) return;
    get(n-1);
    get(n-3);
    printf("%d", n);
}
```

A.

15

B.

25

C.

35

D.

45

Q.27

What will be the output of the following C program?

```
void count(int n)
```

```
{
```

```
static int d = 1;
```

```
1 printf("%d ", n);
```

```
2 printf("%d ", d);
```

```
3 d++;
```

```
4 if(n > 1) count(n-1);
```

```
5 printf("%d ", d);
```

```
}
```

```
int main()
```

```
{
```

```
count(3);
```

```
}
```

A.

3 1 2 2 1 3 4 4 4

B.

3 1 2 1 1 1 2 2 2

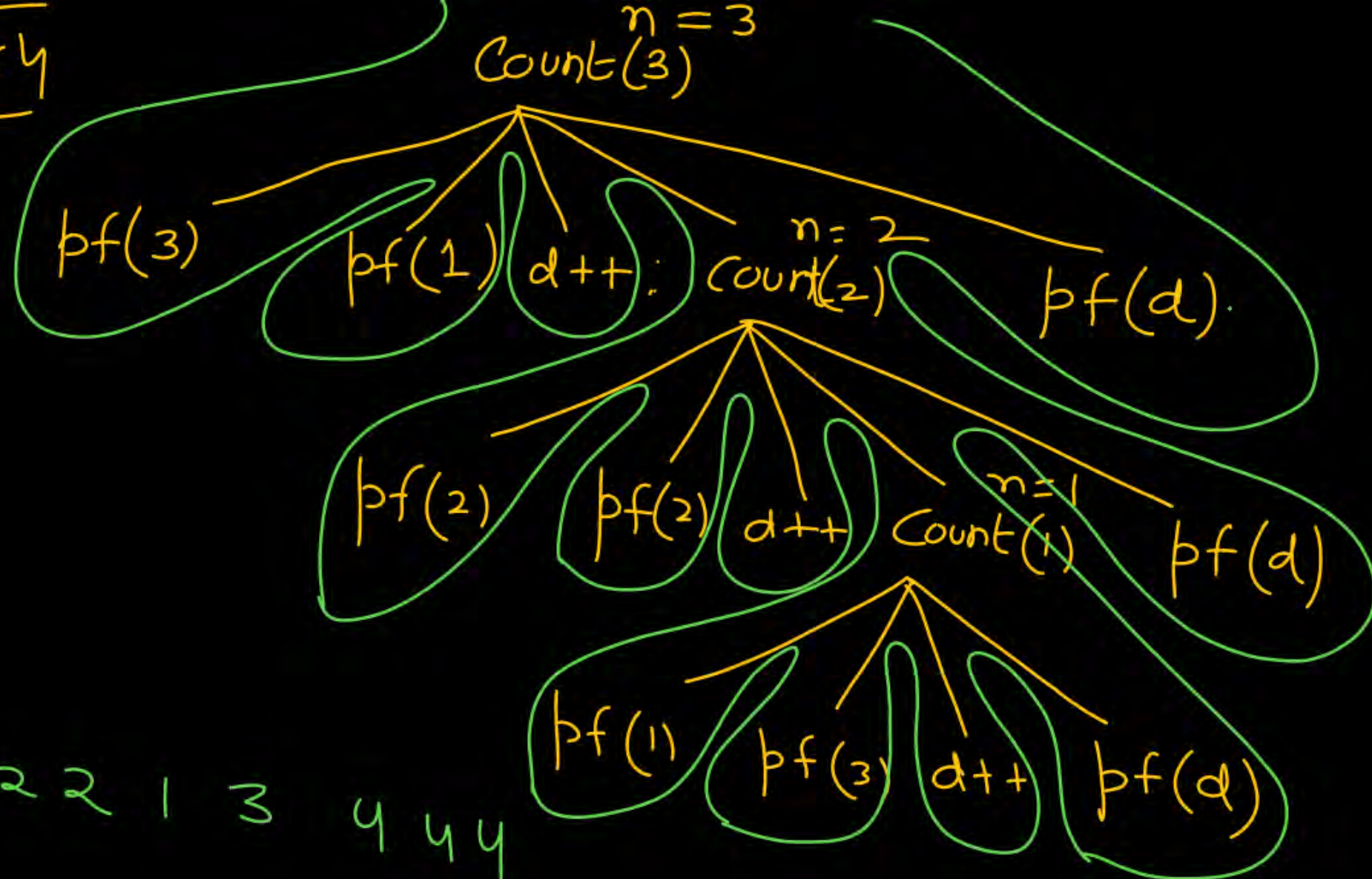
C.

3 1 2 2 1 3 4

D.

3 1 2 1 1 1 2

d 1 2 3 4



3 1 2 2 1 3

4 4 4

Q.28

What will be the output of the C program?

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    function();
```

```
    return 0;
```

```
}
```

```
void function()
```

```
{
```

```
    printf("Function in C is awesome");
```

```
}
```

A.

Function in C is awesome

B.

no output

C.

Runtime error

D.

Compilation error

Q.29

What will be the output of the C program?

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
1  main();
```

```
2  return 0;
```

```
3  }
```

A.

Runtime error ✓✓

C.

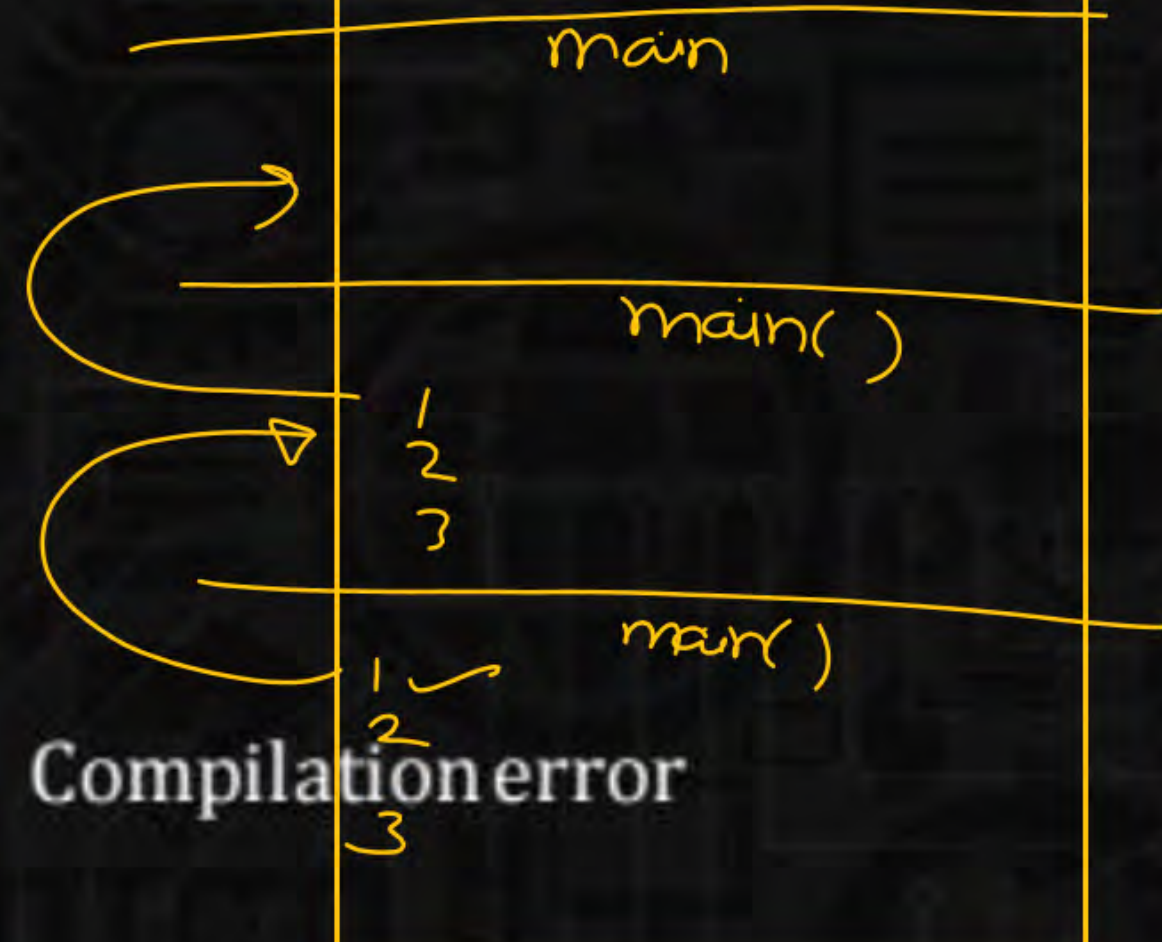
0

B.

Compilation error

D.

None of these



Tower of Hanoi

n distinct plates

$n=3$



A
(Source)



B
(Aux)



C
(dest)

- ① Only 1 disk at a time.
- ② Only allowed to put smaller plate above larger plate.

Recursion

$n=3$ ndisk



Source



Aux



dest

step 1
n-1 disk from
Source to Aux
using dest.



Source



Aux



dest

step 2

Move Source → dest



Source



Aux



dest

if ($n==1$)

Move Source → dest.

else {

Steps:



Source



Aux



dest.

Move (n-1) disk from
Aux to dest
using source.

}

TOH (n, Source, dest, Aux)
{

if (n == 1)

Move Source $\rightarrow$ dest

return ;

else {

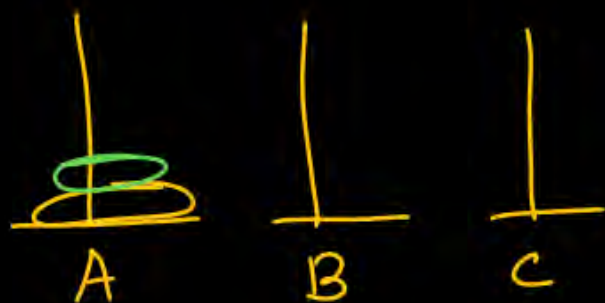
1.) TOH(n-1, Source, Aux, dest);

2.) Move Source $\rightarrow$ dest

3.) TOH(n-1, Aux, dest, Source);
}

}

n=2



TOH (ⁿ Source dest Aux
2, A, C, B)

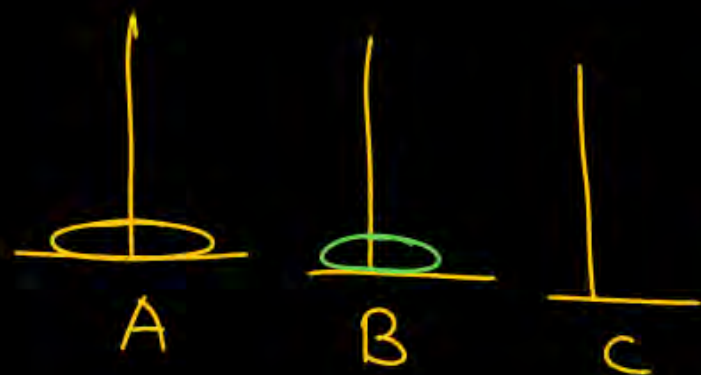
TOH (^{Source dest Aux}
1, A, B, C)

A → C

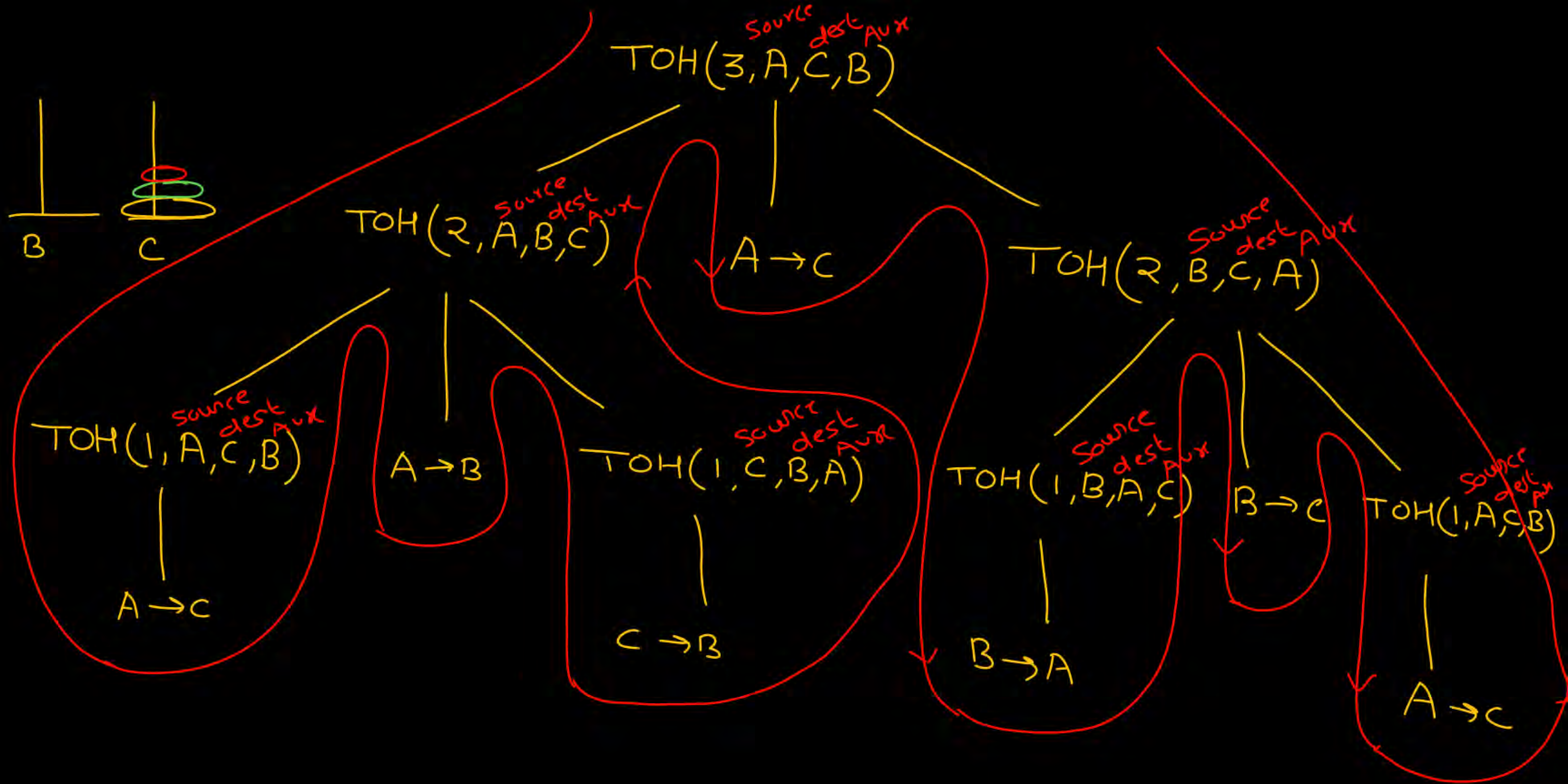
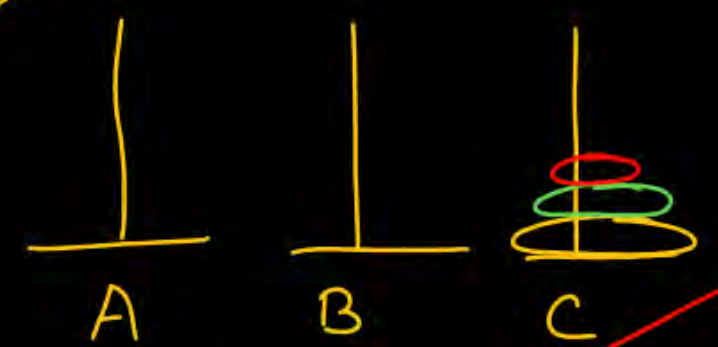
TOH (^{Source dest Aux}
1, B, C, A)

B → C

A → B



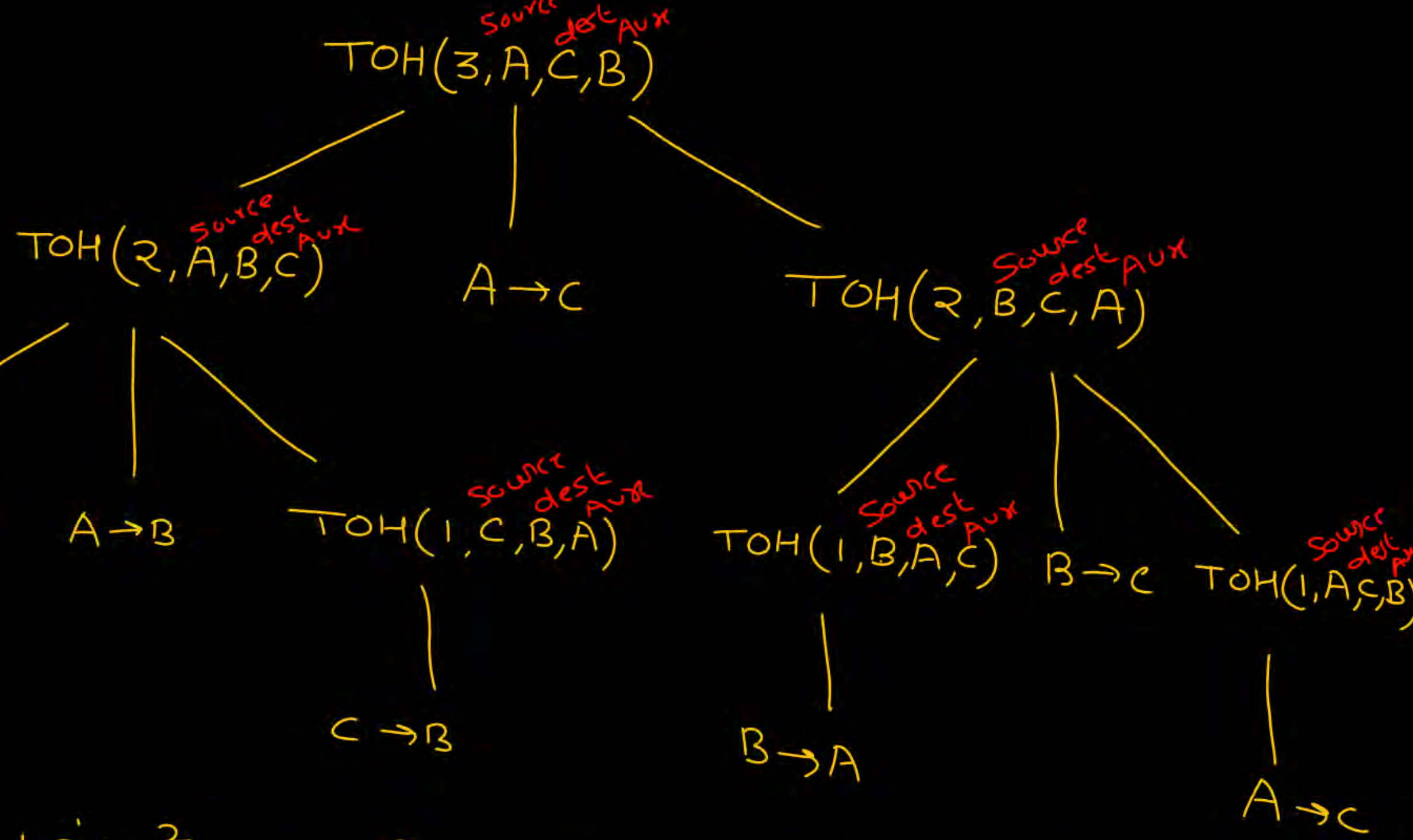
n=3



n=3



①



②

TOH(1, A, C, B)

A → C

④

$$1 + 2 + 2^2$$

$$\Rightarrow \frac{2^3 - 1}{2 - 1} = 2^3 - 1$$

$$1 + 2 + 2^2 + \dots + 2^{n-1}$$
$$\Rightarrow \frac{2^n - 1}{2 - 1} = 2^n - 1$$

Q What is total no. of function calls in $TOH(n, S, d, A)$?

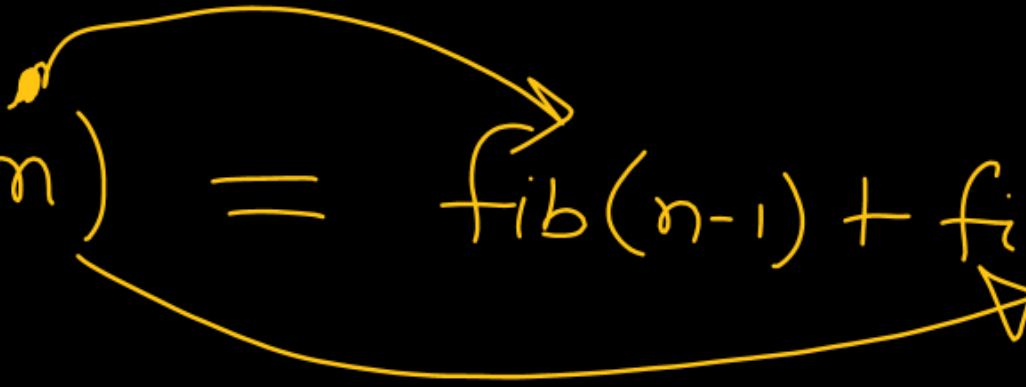
$$2^n$$

Q] n disk $\rightarrow$ total no. of plate movement.

disk	Move
1	1
2	3
3	7

Fibonacci Series

$\text{fib}(0)$ $\text{fib}(1)$ $\text{fib}(2)$
 $\Downarrow$ $\Downarrow$ $\Downarrow$
0, 1, 1, 2, 3, 5, - -

$$\text{fib}(n) = \text{fib}(n-1) + \text{fib}(n-2)$$


Doubts?

1, 2, 3

$$a = 10$$

$$a = 10.28$$

① P_{1239} $\rightarrow$ down

9 PM

② P_{1239} $\rightarrow$



